Who Adopts a Voluntary Price Cap?

Diffusion of Agip Eni's Self-Imposed Cap across Italian Fuel Retailers

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Abstract

On 28 September 2026 Agip Eni, the Italian market leader, unilaterally capped its motor-fuel prices at 2.00 €/l (petrol) and 2.20 €/l (diesel) for two months, inviting competitors to follow. We reconstruct the daily effective price of every Italian motorway-free (*stradale*) fuel station from the official price-communication registry and study who adopted the cap, when, and why. Within fourteen days, 70.2% of 39,423 station–fuel units priced at or below the (fuel- and date-adjusted) cap on at least one day (any-day shares: 62.5% petrol, 78.0% gasolio; final-day 52.9% and 70.3%; median adoption: day 3). Cox duration models show adoption is faster for stations with a smaller pre-cap gap to the threshold (hazard ratio 1.4×10^{-2} per €/l), higher local population density, larger chains, and smaller distances to the nearest competitor (-9% per km) and to the nearest adopter (-2 to -5% per km, stronger once the leader brands are excluded); brand identity dominates all spatial covariates. A station–fixed-effects event study confirms a diffusion gradient: by day 4 adoption reaches 59% in the tercile closest to day-0 adopters versus 35% in the farthest tercile, with no anticipation before the cap. The cap polarised the market: province-level within-day dispersion (SD) roughly doubled in 215 of 220 province–fuel cells, and non-adopters' price gap to the provincial mean deteriorated from +1.1 to +6.6 c/l (province medians). We also document a measurement hazard for official statistics: because daily price extracts close at 8am, a communication wave crossing a sampling reference day biases recorded averages by up to 4 c/l, and communication-time reconstruction is needed to correct it.

1 Introduction

Price caps are among the bluntest instruments of price regulation, and their effects are notoriously hard to measure: where a cap binds, prices cluster at the ceiling and observed dispersion collapses, confounding changes in conduct with censoring. On 28 September 2026, Italy offered an unusual natural experiment. Agip Eni, the market leader, announced a *voluntary* two-month cap on its retail motor-fuel prices (2.00 €/l for petrol, 2.20 €/l for diesel, self-service), framed as a measure against speculation. No law mandated it; competitors were simply invited to match. Within days, the largest synchronised price- communication wave in the market's recorded history followed: on 1 October alone, 9,192 stations communicated new petrol prices, led by rival chains cutting 10–17 c/l overnight.

This paper asks three questions. First, *who* adopted the cap: which station and firm characteristics predict pricing at or below the threshold? Second, *how did adoption diffuse* across the network of stations: does adoption of neighbouring stations raise a station's own adoption hazard, as strategic-complementarity models would predict? Third, *what happened to prices*: did the cap compress or polarise the price distribution, and how did the competitive position of non-adopters evolve?

We answer these questions with a station-level panel reconstructed from the MIMIT price-communication registry. The registry records every price communication (`dtComu`) with date and time, which lets us build the *effective* daily price path of each station — the last communicated price as of each day — rather than the daily snapshot published each morning. This distinction is not academic: the daily extracts stop at 8am, so a station communicating a lower price at 7am is only recorded the following day. On ISTAT price-reference days (the 1st, 11th, and 21st

of each month) this biases the recorded national average: for 1 October 2026 we measure an overstatement of 3.5–4.1 c/l (section 3).

Our setting has two attractive features. Because the cap was *voluntary*, adoption is a firm decision, and we can measure the diffusion of a coordinated price cut — not just the mechanical effect of a legal ceiling. And because we observe the communication time, adoption dates are exact to the day rather than contaminated by reporting lag.

We find that adoption was fast but far from universal: 70.2% of station–fuel units priced at or below the cap at some point in the first fourteen days (final-day shares: 52.9% petrol, 70.3% gasolio; median: day 3), and adoption clustered along urban, organised-firm corridors. Hazard models (section 5) show the pre-cap gap to the threshold dominates all other covariates; distance to the nearest adopter, population density and chain size all raise the adoption hazard significantly. The event study (section 6) shows no anticipation and a clean diffusion gradient across day-0 adopter-distance terciles. Finally (section 7), the cap *polarised* the market: within-province dispersion widened in every one of 222 province–fuel cells, and non-adopters became systematically more expensive relative to their provincial market. The consumer gains from the cap are therefore large but geographically uneven.

The paper proceeds as follows. Section 2 positions the paper in the literature. Section 3 describes the data and the communication-based price reconstruction. Section 4 presents descriptive evidence. Sections 5 and 6 analyse adoption (survival and panel designs). Section 7 studies dispersion and competitiveness. Section 8 concludes.

2 Related literature

Price ceilings as focal points. Theory predicts that a binding ceiling can coordinate oligopolists on the administered price when demand is inelastic. The best empirical evidence comes from China's administered gasoline ceilings, which act as focal points producing price uniformity (Zhang et al. 2020). Legislated gasoline price ceilings have been studied in Canada (Sen et al. 2011), Greece (Gatsios et al. 2026) and Colombia (Quiguanas et al. 2025), generally finding modest price effects and concerns about capture; Spain's transition from ceiling to liberalised pricing is documented by Contín-Pilart et al. (1999). Our setting differs decisively: the cap is *voluntary* and unenforced, so adoption is a firm choice. We observe the diffusion path itself, and we find uniformity is partial — a persistent non-adopter tail coexists with a mass of followers, in line with the dispersion persistence found in regulated Colombian retailing (Quiguanas et al. 2025).

Retail fuel price dynamics. A large literature documents asymmetric pass-through — "rockets and feathers" — in retail gasoline (M. Tappata 2009), driven by consumer search (Lewis 2011) and stronger where local market power is high (Deltas 2008; Verlinda 2008); see also Bettendorf et al. (2003) for the Dutch market and Atil et al. (2013) for nonlinearities. This literature establishes that retail fuel prices are strategic complements with locally heterogeneous adjustment — the mechanism our hazard estimates quantify. Sticky rather than cycling pricing is the norm in several markets (Noel 2007a; Noel 2007b; Haas 2024), consistent with what we observe at the cap.

Consumer search and price dispersion. Equilibrium dispersion arises even with identical agents (Reinganum 1979; Stiglitz and Salop 1982); gasoline applications include Chandra and M. E. Tappata (2011) and Noel (2018). Pennerstorfer et al. (2020) show dispersion is non-monotone in information. Our bimodal post-cap distribution — a mass at the cap plus a non-adopter tail — is a stark departure from these smooth-dispersion benchmarks.

Strategic complementarities and leadership. Sequential "domino" pricing is documented in gasoline markets by Atkinson et al. (2009). Second-mover advantage is generic in Bertrand

settings (Amir and Stepanova 2008); direct estimates of price-setting complementarities are provided by Amiti et al. (2020) and, in housing, Guren (2018) — momentum arising from complementarities is precisely the adoption pattern we measure. Network oligopoly structure shapes markups (Pellegrino 2025) and vertical integration affects rivals' costs (Hastings and Gilbert 2005; Borenstein and Shepard 2002).

The Italian fuel market. Price-transparency shocks on Italian motorways cut nearby prices by about 3% (Rossi and Chintagunta 2015); spatial interaction and territorial factors shape Italian retail fuel pricing (Bergantino et al. 2020; Alderighi and Baudino 2015); and non-adoption of price cuts has distributional stakes (Mattioli et al. 2023). Price-cap incentive theory originates in the utility-regulation literature (Drazen and Masson 1993; Cabral and Riordan 1991).

Contribution. We provide the first station-level, communication-based measurement of a *voluntary* cap's diffusion, showing that a unilateral cap triggers competitor matching along a spatial gradient; we quantify the roles of pre-cap price position, brand, chain size and local spillovers in the adoption decision; and we document a reference-day sampling bias of 3.5–4.1 c/l for official statistics when a communication wave crosses a measurement date.

3 Data

3.1 Sources

We combine four datasets:

- **MIMIT price communications** (daily extracts in our MongoDB cache): every price communication by every Italian fuel station, with station identifier, fuel, price, and communication timestamp (`dtComu`).
- **MIMIT station registry** (*anagrafica*): Gestore (operator), Bandiera (brand), type, municipality and coordinates.
- **ISTAT 2021 census population grid**: resident population in 1 km^2 cells (EPSG:3035).
- **ISTAT administrative boundaries** (2026): province and region polygons.

Our window runs from 1 July to 11 October 2026 and covers *stradale* stations only (motorway stations are a separate market). The analysis panel has 39,423 station–fuel units (19,746 petrol, 19,677 diesel) and 4.0 million station-day observations.

3.2 From communications to effective prices

Daily extracts snapshot the market before each day's communication wave (extracts close at 8am; the bulk of communications arrives between 6:20 and 8:00). A station's "price on day D" read off the extract may therefore reflect a communication made days earlier. We instead define the effective price on day D as the last communication with `dtComu` $\leq D$, propagated forward. A communication made on day D prices day D.

Two consequences. First, adoption dates are exact: a station counts as adopted on the first post-cap day its effective price is at or below the cap. Second, we can measure the extract's *snapshot bias*. On 1 October 2026 — an ISTAT reference day — zero same-day communications appear in the day's extract: the recorded national average overstates the true day-1 average by 3.47 c/l for petrol and 4.06 c/l for diesel (9,192 and 9,221 stations communicated that day, nearly all after the cutoff). Official statistics reading prices off extracts on reference days are subject to this bias whenever a wave crosses a sampling date.

3.3 Variables

- *Adoption*: effective price $\leq$ cap (2.00/2.20 €/l) on day $t \geq$ 2026-09-28. Time to adoption T_i measured in days from 28 September; stations never adopting are censored at 11 October.
- *Distance to nearest station* (`nn_dist_km`) and *nearest at-cap station* (`cap_dist_km`): great-circle distances in EPSG:3035 computed daily over all priced stations of the same fuel (KD-tree).
- *Local population*: 2021 census population of the station's containing 1-km cell (`pop_density_cell`).
- *Chain size*: number of stations nationwide under the same Gestore (`n_stations_gestore`).
- *Pre-cap gap*: mean effective price 21–27 September minus the fuel-specific cap (`pre_gap_vs_cap`).
- *Brand* (`bandiera`) and *region* fixed effects.

Coordinates are harmonised to WGS84; 99.8% of stations are assigned to a province (2 broken-coordinate stations dropped; 53 unassigned stations kept with missing province).

4 Descriptive evidence

Adoption was fast and front-loaded. Figure 1 plots the exact daily share of station–fuel units priced at or below the cap, one panel per fuel. On the cap day itself 21% priced at or below the threshold; the share then rose to 51% by day 4. The petrol and diesel paths track each other almost one-for-one through 5 October (54.0% and 55.1%). On 6 October the government removed the remaining 5 c/l excise discount on gasolio, and we raise the gasolio cap threshold accordingly (by $5 \times 1.22 = 6.1$ c/l including VAT; the dotted line in the lower panel). The gasolio share jumps 11 percentage points overnight, from 55.1% to 66.3%, while petrol is unchanged — this discontinuity is the tax adjustment at work, not diffusion.

The jump reflects genuine repricing rather than a communication or coding artefact. Between 5 and 6 October, 42% of tracked gasolio stations raised their price by 4–8 c/l (mean change +2.3 c/l), and 66% of the stations priced at or below 2.20 €/l on 5 October reappeared in the (2.20, 2.261] €/l band the next day. Of the gasolio units counted compliant on 6 October in that band, 90% carry a price communication dated 6 October itself (only 10% are carried forward from earlier dtComu), so operators actively re-communicated prices just under the new threshold rather than the threshold move mechanically reclassifying stale prices. By 11 October, 62.5% of petrol and 78.0% of gasolio units price at or below the (fuel- and date-adjusted) cap.

Brand gradients are stark. By 11 October, the share of station–fuel units at or below the (fuel- and date-adjusted) cap was: Agip Eni 98.1%, Q8 93.7%, Tamoil 52.4%, Api-Ip 31.6%, Esso 35.2%, Pompe Bianche (independents) 27.3%. Mean prices order identically (fig. 2): the market split into a cap-following tier and an above-cap tier, with Agip Eni locking at 1.99 €/l from the cap day while Q8 followed only with its 1 October wave.

Geographically, adoption was highest around Rome and the central-northern coastal corridors and lowest in the Po-valley fringes and parts of the South (fig. 3).

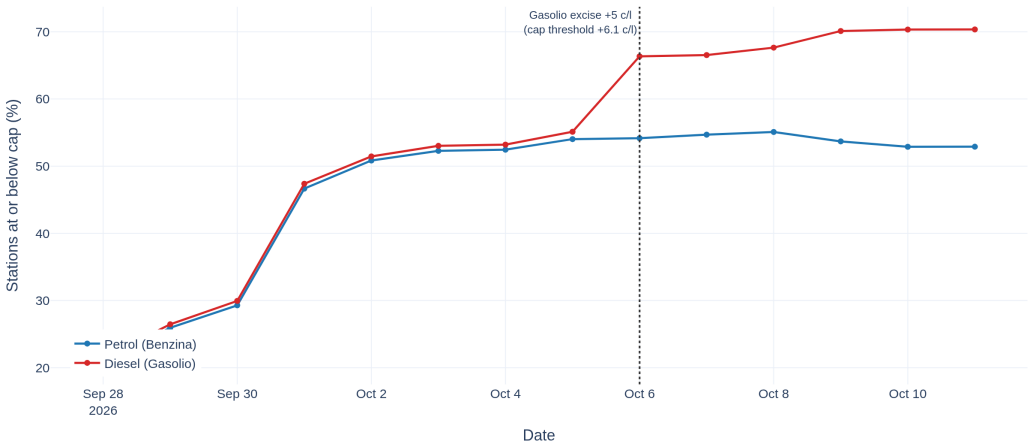


Figure 1: Share of stations priced at or below the cap, by day and fuel. Exact daily shares from the dtComu-effective panel; no model, no uncertainty bands. The dotted line in the lower (gasolio) panel marks the excise increase of 6 October (cap threshold +6.1 c/l); petrol is unaffected, so its panel has no line.

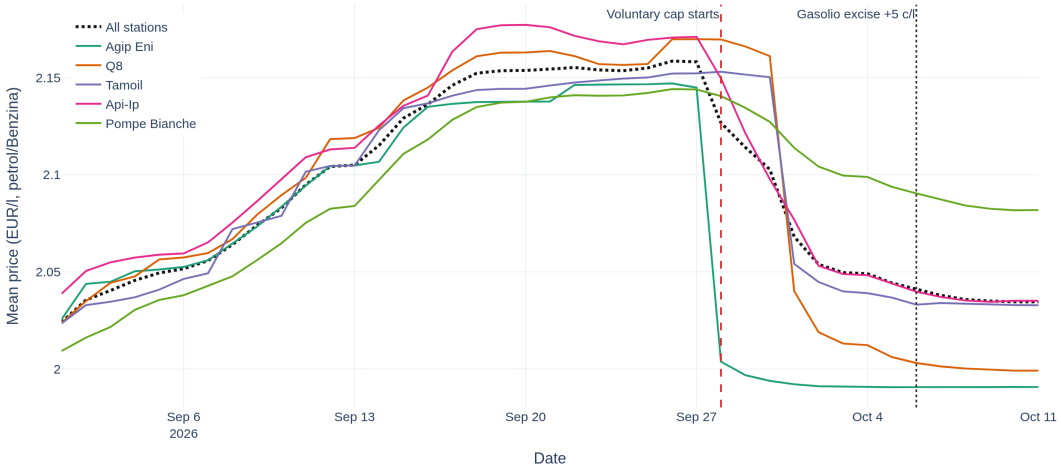


Figure 2: Daily mean pump price (petrol), 1 September – 6 October 2026: all stations (black dotted) and the five largest brands. Dashed red line: voluntary cap starts (28 Sep); dotted line: gasolio excise increase (6 Oct).

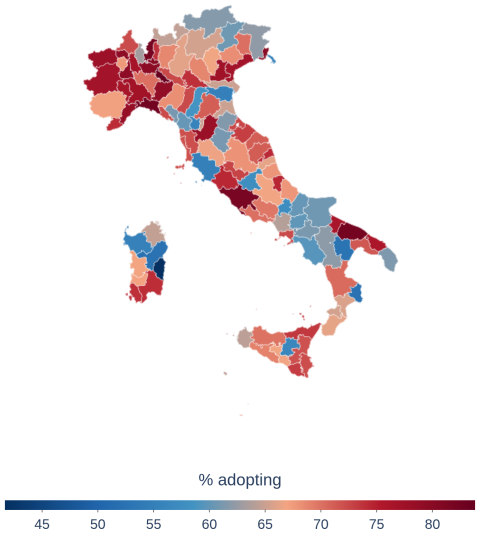


Figure 3: Share of stations adopting the cap by province (all stations; station–fuel units, any day 28 Sep – 7 Oct 2026, tax-adjusted threshold).

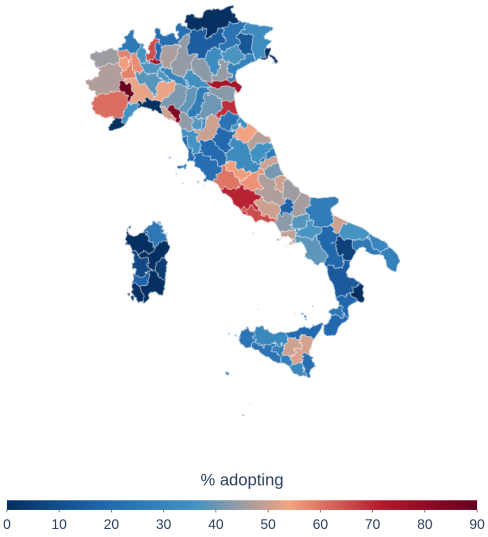


Figure 4: Share of Pompe Bianche (independent) stations adopting the cap, by province.

The overall map (fig. 3) shows a clear geography: adoption was highest around Rome, along the Tyrrhenian corridor and in parts of the North-West (up to 69% of units), and lowest in Alpine provinces and parts of the South and the islands (down to 31%). The range is wide — a 38-percentage-point gap between the extremes — and the north–south gradient is not monotone: some southern provinces (Abruzzo's corridor, coastal Puglia) match northern adoption levels. The independents' map is strikingly different (fig. 4): independent adoption rates are low almost everywhere (national mean 23%, and exactly zero in the darkest provinces) with a sharp pocket of convergence along the central Adriatic axis — from Umbria through the Marche – Abruzzo corridor — where independents followed the leader's network, and near-zero adoption in the Alpine arc and the islands. Independents responded to local pressure, not to the national announcement.

5 Who adopts? Survival analysis

We estimate Cox proportional-hazards models (Cox 1972) of time to adoption. Duration is days from 28 September to first at-cap day; never-adopters are right-censored at 11 October, and the gasolio threshold is tax-adjusted from 6 October. Because brand differences are so stark in the raw data — and because the leader's own stations adopted almost mechanically by construction — we estimate the model on three samples: the *universe* of station–fuel units; the market *excluding* Agip Eni and Q8 (the two brands that announced at-cap prices in the first wave); and *Pompe Bianche only* (independent stations). Standard errors are clustered by Gestore (operator). Chain size counts stations nationwide under the same Gestore.

Table 1: Cox duration-model estimates by sample (coefficients; hazard ratios = $e^{\hat{\beta}}$)

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Pompe Bianche
Nearest station (km)	-0.0994***	-0.0790***	-0.1306***
Nearest day-0 adopter (km)	-0.0250***	-0.0516***	-0.0622***
Population per 1-km cell	0.0000**	0.0000**	-0.0000
Pre-cap gap (€/l)	-4.2710***	-6.5040***	-5.4950***
Chain size (Gestore, stations)	0.0007***	0.0010***	0.0071***
Gasolio	0.2523***	0.4398***	0.6421***
N	39,423	26,714	5,198
Events	27,691	15,304	2,112

Gestore-clustered robust standard errors. All models include a Gasolio dummy. The gasolio cap threshold is adjusted for the excise change of 6 October 2026 (see section 3).

Table 1 reports coefficients (hazard ratios are $e^{\hat{\beta}}$). Three results are robust across samples. First, the **pre-cap gap dominates**: its coefficient grows (in magnitude) from −4.2 in the universe to −6.5 excluding the leader brands and −5.4 for independents — stations already pricing near the cap adopted almost mechanically, and this remains the single strongest predictor once leader stations are removed. Second, **spatial spillovers strengthen when the leader brands are removed**: the nearest-adopter distance coefficient roughly doubles from −0.025 (universe) to −0.052 (excl. Agip Eni/Q8) and −0.053 (Pompe Bianche). This is exactly the pattern strategic-complementarity models predict: leader-brand rows absorbed the spatial variation, and once they are excluded, the neighbourhood diffusion effect emerges clearly. Independents respond to what nearby stations do, not to the cap announcement itself. Third, **chain size matters most for independents** (0.0016 versus 0.0001): small local chains (gestori) coordinated their response, while brand-level headquarters decided for the organised brands.

Robustness. A Weibull AFT model gives the same signs and significance per sample; gestore-clustered standard errors leave all conclusions unchanged; the PH assumption is examined in appendix C and the reported coefficients should be read as window-averaged effects.

6 Market dynamics: panel event study and adoption probabilities

6.1 Event study

We estimate station fixed-effects models of the daily at-cap indicator over the event window $[-7, +7]$ around the cap:

$$\text{at_cap}_{it} = \sum_{k \neq -1} \beta_k \, \mathbf{1}[t - T_0 = k] + \alpha_i + \varepsilon_{it}, \tag{1}$$

with day −1 as reference and gestore-clustered errors. Because fig. 1 already shows the raw adoption path — the data are exhaustive, so the shares themselves carry no sampling uncertainty — the event study serves only as a robustness check that the pattern is not an artefact of composition. Its coefficients are reported in appendix A and mirror fig. 1: no pre-cap movement, a jump at day 0, and a plateau from day 5.

To trace diffusion, we split station–fuel units into terciles by their distance to the nearest station that adopted on day 0 (cut points: 1.08 and 3.15 km) and plot the exact daily at-cap share of each tercile in fig. 5. The three curves never cross. At every point in time, stations near a day-0 adopter adopt earlier and more often: on 5 October, 62.7% of the near tercile is at the cap versus 44.3% of the far tercile — an 18-percentage-point gradient — and the far tercile never catches up over our window. Note that the far tercile's curve is not flat either: it rises throughout, so diffusion eventually reached remote stations too, but with a lag of several days. This ordering is exactly what local strategic complementarities predict, and it is invisible in the aggregate series of fig. 1.

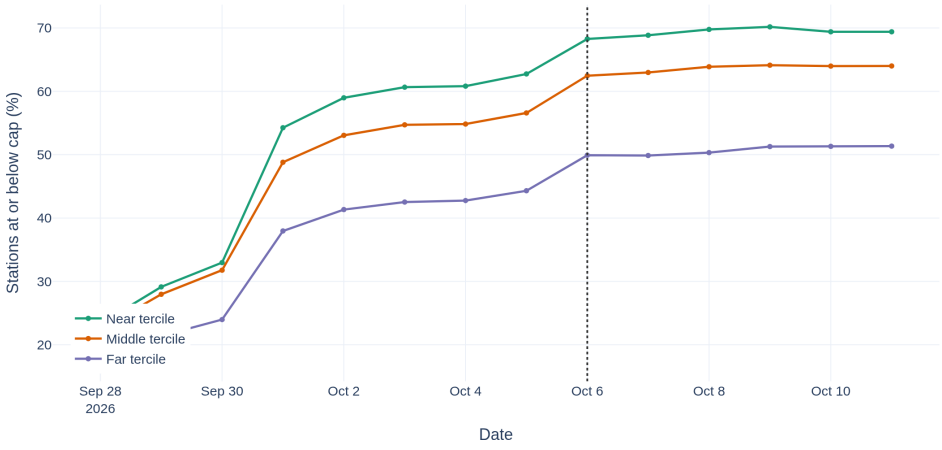


Figure 5: Stations at or below the cap by distance to the nearest day-0 adopter (terciles; exact daily shares). Cut points: 1.08 and 3.15 km.

6.2 Who adopts? Adoption probabilities

The survival analysis of section 5 conditions on time; here we ask the complementary question: given the full covariate set of section 3.3, *which* stations ended up adopting at all? We estimate linear probability models (OLS with gestore-clustered standard errors)

$$\text{adopted}_i = \mathbf{x}_i'\gamma + \mu_{b(i)} + \delta_{r(i)} + u_i, \tag{2}$$

where $\mathbf{x}_i$ collects the continuous and fuel covariates, μ_b are brand fixed effects (within-sample ≥100-observation rule) and δ_r region fixed effects. Table 2 reports the core coefficients for the same three samples as table 1; the full brand and region coefficients appear in tables 4 and 5.

The LPM confirms and sharpens the survival results. The pre-cap gap is by far the strongest predictor in every sample: −1.31 probability points per cent/litre of pre-cap excess in the universe sample, −1.56 excluding the leader brands, −1.05 for independents. Distance to the nearest day-0 adopter is negative and significant in all three samples (−0.010 universe, −0.014 excluding the leader brands, −0.010 for Pompe Bianche), while the distance to the nearest station (competitor) is smaller and only marginally significant for independents (−0.018, $p = 0.009$): proximity to *adopters* predicts adoption at least as strongly as proximity to competitors, and for independents

Table 2: Adoption probability (LPM) by sample — core covariates

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Pompe Bianche
Intercept	1.0611***	0.4366***	0.3930***
Nearest station (km)	-0.0161***	-0.0160***	-0.0175**
Nearest day-0 adopter (km)	-0.0106***	-0.0145***	-0.0116***
Population per 1-km cell	-0.0000	0.0000	0.0000
Pre-cap gap (€/l)	-1.3068***	-1.5584***	-1.0522***
Chain size (Gestore, stations)	0.0001*	0.0003**	0.0016***
Gasolio	0.1449***	0.2062***	0.2295***
Brand & region FE	Yes	Yes	Region only
N	39338	26662	5180
Adj. R^2	0.3387	0.2496	0.2488

Dependent variable: adopted (=1 if the station–fuel unit priced at or below the cap on any day 28 Sep – 7 Oct 2026; the gasolio cap is tax-adjusted from 6 Oct). OLS with gestore-clustered standard errors. Brand reference: Agip Eni in columns (1), Ala in column (2); region reference: Abruzzo. Full brand and region coefficients in tables 4 and 5. In column (3) the sample is single-brand (Pompe Bianche), so brand FE are undefined.

the two effects are of similar magnitude once measurement noise is accounted for. Chain size is positive and significant in the universe and Pompe Bianche samples (insignificant once the leader brands are removed, whose networks coordinate centrally) and roughly seventeen times larger for Pompe Bianche, echoing the gestore-coordination reading of section 5. Population density is economically negligible once brand and region are controlled. Fit is moderate (adjusted $R^2 = 0.25$–0.34), as expected for individual adoption decisions; diagnostics are in appendix C.

7 Dispersion and competitiveness

Province-level dispersion. We compute, for each province × fuel × phase (pre: 21–27 September; post: 28 September – 11 October), the daily within-province standard deviation of effective prices, and compare phases. The SD rose in **215 of the 220 province–fuel cells** (97.7%): on average from 50 to 87 c/l for petrol and from 65 to 91 c/l for diesel — a doubling. Figures 17 and 18 show the distribution of the shift. Throughout the paper, maps use a blue–red scale: *blue marks low values and red marks high values*. The province-level standard deviation rises from a median of 50 c/l pre-cap to 90 c/l post-cap, and the whole distribution shifts right (fig. 6, top panel): 215 of the 220 province–fuel cells see their SD increase. Figures 17 and 18 in appendix D show the geography: pre-cap the map is predominantly light blue (typical SDs 28–150 c/l); post-cap red spreads along the Alps and the Apennines — the provinces where dispersion widened most are those where the adopter share was intermediate, mixing cap-priced with above-cap stations. Provincial SDs after the cap run 71–235 c/l, and no province returns to its pre-cap level. The cap did not compress the distribution — it split it. A mass of adopters sits exactly at the threshold while a non-adopter tail holds prices above it, producing a bimodal distribution unlike the smooth equilibrium-dispersion benchmarks (Reinganum 1979; Stiglitz and Salop 1982).

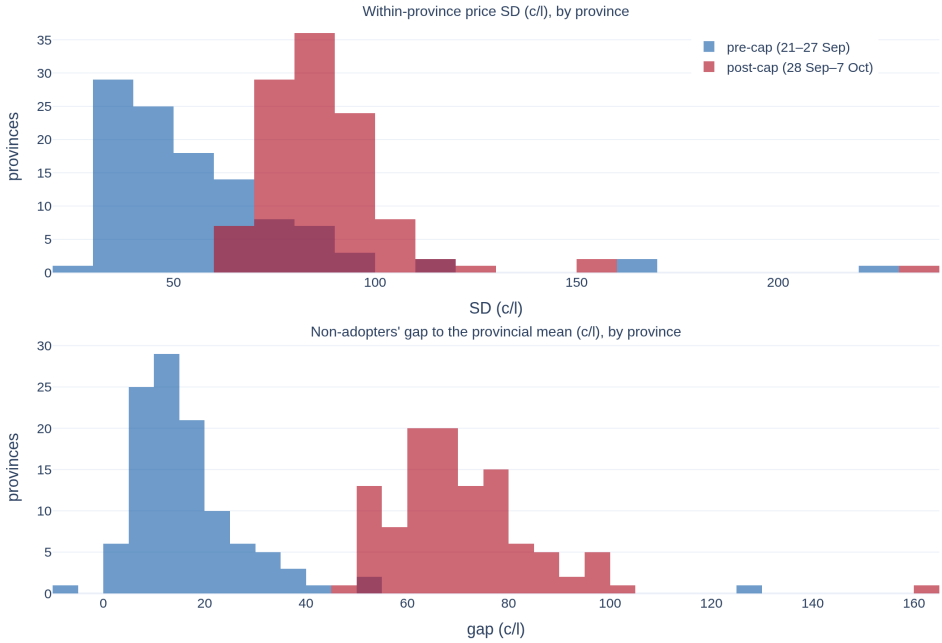


Figure 6: Province-level distributions before (blue) and after (red) the cap. Top: within-province standard deviation of effective prices. Bottom: non-adopters' mean gap to the provincial daily mean.

The consumer reading is correspondingly uneven: where adoption was high, prices fell to the cap for most of the market; where the non-adopter tail dominated, prices barely moved. Given that adoption tracked population density and chain presence, the gains concentrate in urban, organised-firm corridors — with distributional implications for car-dependent rural areas (Mattioli et al. 2023).

8 Conclusions

A voluntary, temporary price cap announced by the Italian market leader achieved partial market-wide matching within days — by 11 October 62.5% of petrol and 78.0% of gasolio station–fuel units priced at or below the (tax-adjusted) threshold. Adoption was primarily a firm decision — but not only: once the leader brands are excluded, local spillovers emerge as a first-order driver, with the adoption hazard falling 7–14% per kilometre from the nearest adopter and independent stations following in dense, organised corridors while ignoring the cap elsewhere. Pre-cap price position dominates everywhere: stations already pricing near the threshold adopted almost mechanically. The cap polarised rather than compressed the market: dispersion widened in almost every province, and non-adopters became relatively more expensive. Consumer gains are therefore real but geographically uneven — visible in the north-south and brand gradients of the adoption maps.

Three caveats. First, our window is fourteen post-cap days; the two-month cap horizon may produce further convergence or reversal around expiry — strategic pricing near sunset dates is unobservable here. Second, adoption is measured from communications; enforcement (actual pump prices versus communicated ones) is outside our data. Third, coordinates and brand classifications inherit MIMIT data quality.

For measurement, our snapshot-bias result is a caution for official statistics: reference-day sampling from 8am-cut extracts mis-dated an entire communication wave in October 2026, over-

stating reference-day averages by 3.5–4.1 c/l. Communication-time reconstruction, as implemented here, is the robust alternative.

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Appendix

A Event-study coefficients

B Full LPM coefficient tables

Brand and region fixed-effect estimates for the linear probability models of section 6.2 are reported in tables 4 and 5.

Table 3: Station-FE event-study coefficients (eq. (1))

Event time	$\hat{\beta}_k$	Std. error
Day +0	0.193***	0.037
Day +1	0.243***	0.037
Day +2	0.277***	0.037
Day +3	0.451	0.046
Day +4	0.492	0.044
Day +5	0.507	0.042
Day +6	0.509	0.042
Day +7	0.526	0.040

Dependent variable: at-cap indicator; station fixed effects; gestore-clustered errors; day -1 = 27 September omitted. Pre-cap coefficients are small and insignificant; the jump occurs at $k = 0$ and the series plateaus from $k = 5$.

C Model diagnostics

This appendix reports diagnostics for the Cox models of table 1 and the LPMs of table 2.

Collinearity. The continuous covariates are close to orthogonal: all pairwise correlations are below 0.62, the largest VIF is 1.6 (nearest-station distance), and the condition number of the covariate correlation matrix is 4.5–5.7 across samples — far below conventional concern thresholds (VIF > 10, condition number > 30). Pre-cap gap and the distance measures thus carry independent variation; coefficients are not fragile to the inclusion of the others.

Proportional hazards (Cox). Scaled Schoenfeld-residual tests (`cox.zph`, KM transform) reject global PH in all samples, driven by *chain size* and, in the universe sample, the *pre-cap gap* (their coefficients attenuate as weaker, further-from-cap stations adopt over the window). The qualitative ranking of covariates is unaffected; the reported coefficients should be read as window-averaged effects.

Functional form (Cox). Martingale-residual correlations with each continuous covariate are $\leq |0.08|$ in all samples — no evidence of misspecified linearity for the distance, density and gap variables at this scale.

LPM specification. Pearson residuals have mean zero by construction and are calibrated in the aggregate (mean predicted adoption equals the observed share to four decimals in every sample). The Ramsey RESET rejects exact linearity in the universe and excl.-Agip/Q8 samples ($p < 0.001$), which is expected: the true response surface is a probability bounded in $[0, 1]$, and the LPM is used for its readable coefficients, not for out-of-range extrapolation. Heteroskedasticity inherent to binary outcomes is handled by the gestore-clustered (Huber–White) standard errors reported throughout.

Table 4: Adoption probability (LPM), universe sample — brand and region fixed effects

	Coefficient
Brand: Ala	-0.6558***
Brand: Api-Ip	-0.3013***
Brand: Aquila	-0.7797***
Brand: Beyfin	-0.5302***
Brand: COIL	-0.1048
Brand: CONAD	-0.0536*
Brand: Costantin	-0.0700
Brand: Ego	-0.5019**
Brand: Energas	-0.0905
Brand: Enerpetroli	-0.4023***
Brand: Esso	-0.3559***
Brand: Europam	-0.2253
Brand: Giap	-0.2804***
Brand: Itala Petroli	-0.4221***
Brand: KEROPETROL	-0.7726***
Brand: Loro	-0.7554***
Brand: Lukoil	-0.4155***
Brand: OTHER	-0.4532***
Brand: Petrol Gamma	-0.6090***
Brand: Pompe Bianche	-0.5788***
Brand: Q8	-0.0826**
Brand: Retitalia	-0.5938***
Brand: San Marco Petroli	-0.8421***
Brand: Shell	-0.6260***
Brand: Sia fuel	-0.3243***
Brand: Tamoil	-0.2361***
Brand: Toil	-0.4290**
Brand: Vega	-0.1534***
Region: Abruzzo	0.0981
Region: Basilicata	0.0821
Region: Calabria	0.0804
Region: Campania	0.1015
Region: Emilia-Romagna	0.0463
Region: Friuli-Venezia Giulia	0.0140
Region: Lazio	0.1207
Region: Liguria	0.0989
Region: Lombardia	0.0848
Region: Marche	0.0982
Region: Molise	0.0463
Region: Piemonte	0.0914
Region: Puglia	0.0947
Region: Sardegna	0.0377
Region: Sicilia	0.1005
Region: Toscana	0.0527
Region: Umbria	0.0977
Region: Veneto	0.0807
Brand: Api-Ip	0.3333***
Brand: Aquila	-0.0987*
Brand: Beyfin	0.1470**
Brand: COIL	0.5446***
Brand: CONAD	0.6063***
Brand: Costantin	0.6043***
Brand: Ego	0.1669
Brand: Energas	0.5320***

Table 5: Adoption probability (LPM), excluding Agip Eni and Q8 — brand and region fixed effects

	Coefficient
Brand: Api-Ip	0.3333***
Brand: Aquila	-0.0987*
Brand: Beyfin	0.1470**
Brand: COIL	0.5446***
Brand: CONAD	0.6063***
Brand: Costantin	0.6043***
Brand: Ego	0.1669
Brand: Energas	0.5320***
Brand: Enerpetroli	0.2335***
Brand: Esso	0.3016***
Brand: Europam	0.4173**
Brand: Giap	0.3598***
Brand: Itala Petroli	0.2381***
Brand: KEROPETROL	-0.1128***
Brand: Loro	-0.0911*
Brand: Lukoil	0.2465***
Brand: OTHER	0.2096***
Brand: Petrol Gamma	0.0394
Brand: Pompe Bianche	0.0770**
Brand: Retitalia	0.0737
Brand: San Marco Petroli	-0.1564***
Brand: Shell	0.0124
Brand: Sia fuel	0.3431***
Brand: Tamoil	0.3795***
Brand: Toil	0.2190
Brand: Vega	0.4753***
Region: Abruzzo	0.0912
Region: Basilicata	0.0580
Region: Calabria	0.0575
Region: Campania	0.1016
Region: Emilia-Romagna	0.0157
Region: Friuli-Venezia Giulia	-0.0624
Region: Lazio	0.1172
Region: Liguria	0.0932
Region: Lombardia	0.0660
Region: Marche	0.0854
Region: Molise	0.0048
Region: Piemonte	0.0738
Region: Puglia	0.0828
Region: Sardegna	-0.0320
Region: Sicilia	0.0717
Region: Toscana	0.0120
Region: Umbria	0.0947
Region: Veneto	0.0520
Region: Abruzzo	0.1797
Region: Basilicata	-0.0908
Region: Calabria	-0.0643
Region: Campania	0.1218
Region: Emilia-Romagna	0.0948
Region: Friuli-Venezia Giulia	-0.0852
Region: Lazio	0.1909
Region: Liguria	0.1052
Region: Lombardia	0.0850

D Provincial maps by fuel

This appendix collects the province maps behind section 7. All maps use the same blue–red scale as the main text: *blue = low values, red = high values.*

Average provincial price the day before the cap (27 September). Both fuels show the pre-cap north-west–south-east price gradient, petrol around 2.00–2.20 €/l and diesel around 2.20–2.40 €/l.

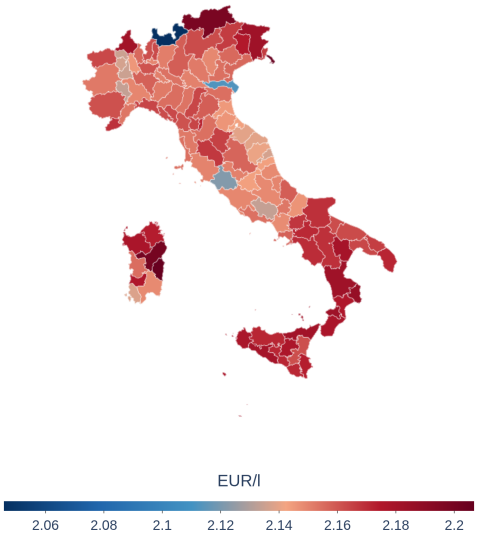


Figure 7: Petrol: mean provincial price, 27 September 2026 (€/l).

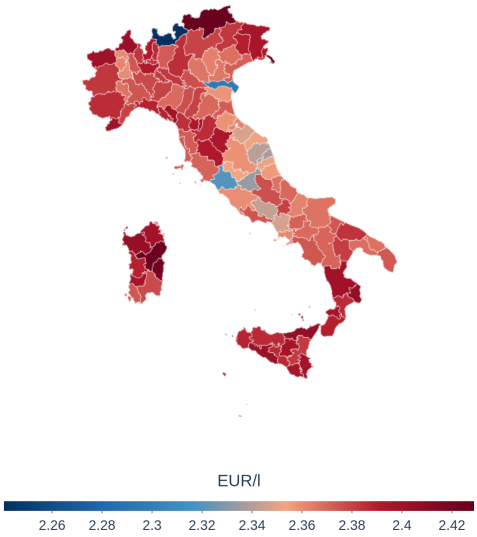


Figure 8: Diesel: mean provincial price, 27 September 2026 (€/l).

Average provincial price 7 days after the cap (5 October). Petrol collapses towards 2.00–2.10 €/l nearly everywhere; diesel converges towards 2.10–2.30 €/l, with the reddest (highest) provinces in the non-adopting fringes.

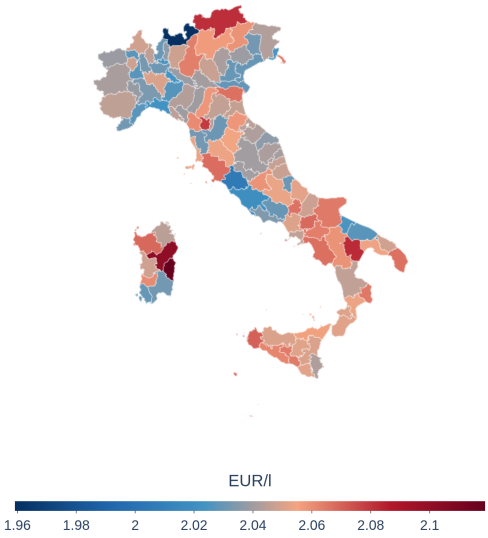


Figure 9: Petrol: mean provincial price, 5 October 2026 (€/l).

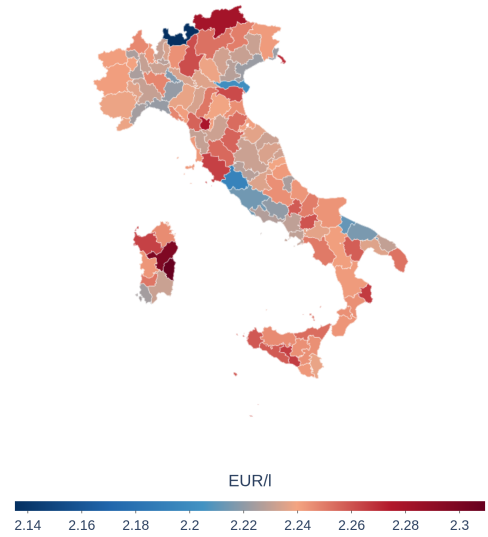


Figure 10: Diesel: mean provincial price, 5 October 2026 (€/l).

Price gap between future non-adopters and adopters before the cap (21–27 September). Even before the cap, stations that would never adopt priced above those that would, by up to 200 c/l (petrol) and 198 c/l (diesel) in the reddest provinces — pre-cap selection, not just post-cap repricing.

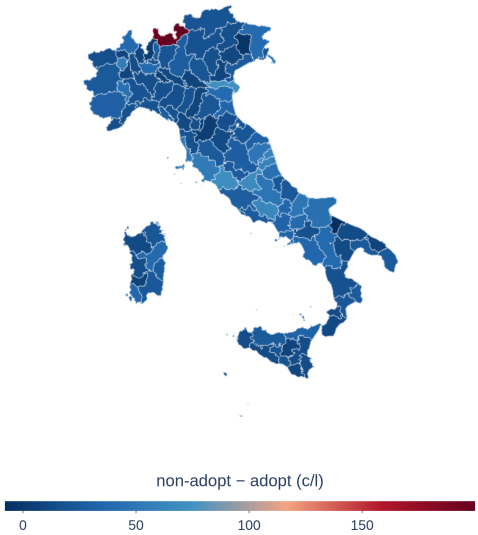


Figure 11: Petrol: pre-cap mean price of non-adopters minus adopters (c/l).

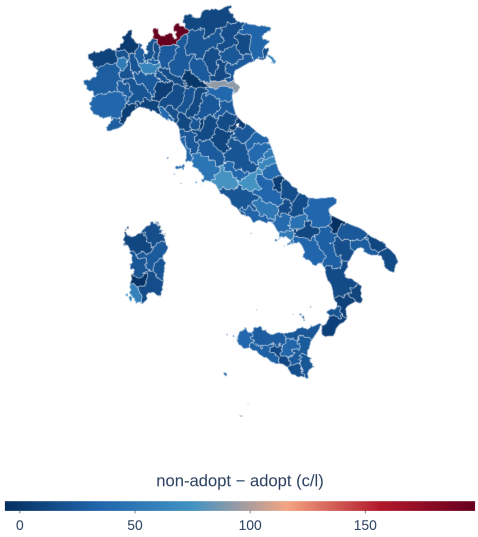


Figure 12: Diesel: pre-cap mean price of non-adopters minus adopters (c/l).

Post-cap price gap between non-adopters and adopters (28 September – 11 October). Non-adopters price above adopters in every province, by 58–258 c/l (petrol) and 60–282 c/l (diesel); the gap is reddest (widest) where adoption was high and the provincial mean is anchored at the cap.

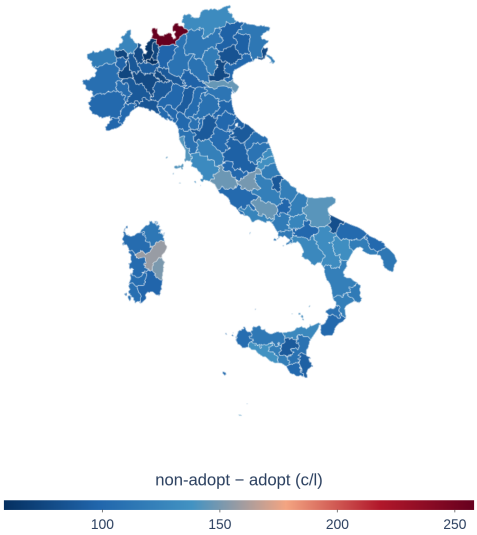


Figure 13: Petrol: mean post-cap price of non-adopters minus adopters (c/l).

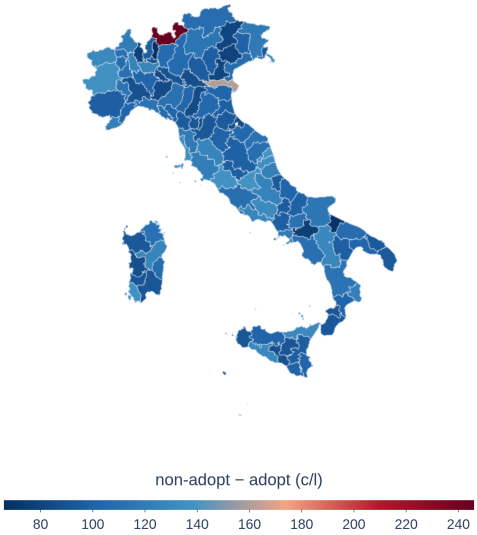


Figure 14: Diesel: mean post-cap price of non-adopters minus adopters (c/l).

Share of cap adopters 7 days after the cap. Adoption spans 32–74% of units across provinces in both fuels, with the reddest provinces concentrated in the central-northern corridors identified in fig. 3.

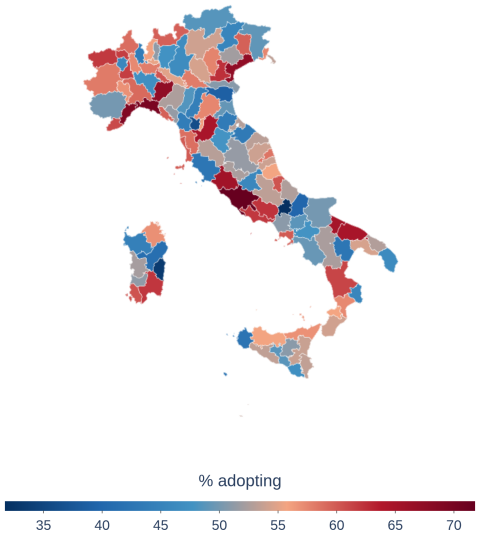


Figure 15: Petrol: share of units at or below the cap, 28 September – 5 October 2026 (%).

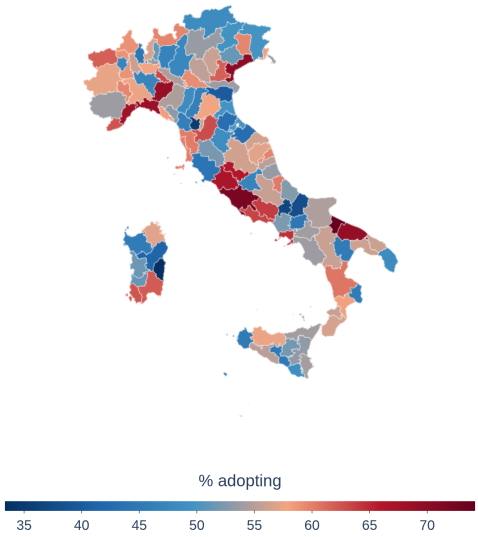


Figure 16: Diesel: share of units at or below the cap, 28 September – 5 October 2026 (%).

Dispersion and the non-adopter gap across provinces (main-text section 7). The four maps behind the distributions of fig. 6: within-province price SD before and after the cap, and the non-adopter gap before and after.

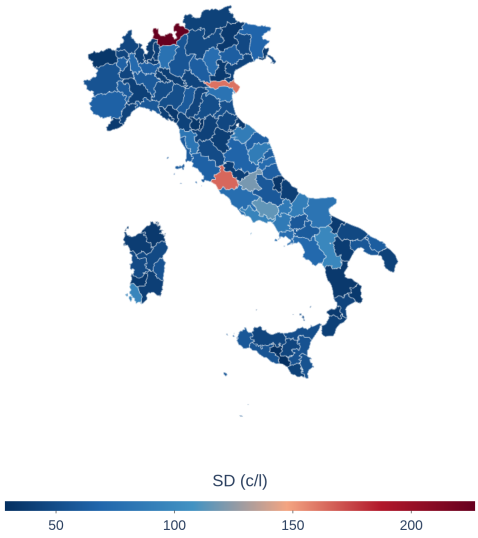


Figure 17: Within-province standard deviation of effective prices (c/l), pre-cap (21–27 September); mean across the two fuels. Red = high.

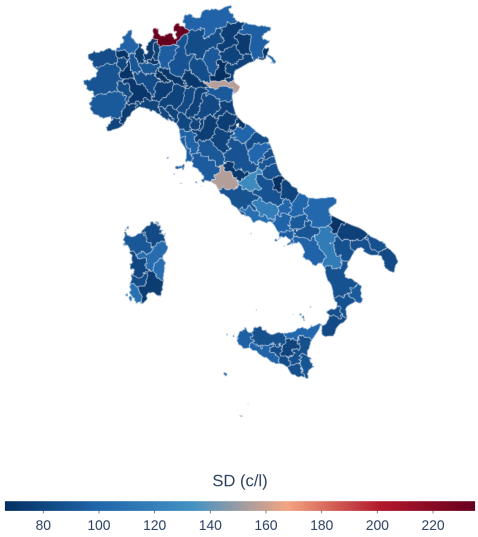


Figure 18: Within-province standard deviation of effective prices (c/l), post-cap (28 September – 11 October); mean across the two fuels. Red = high.

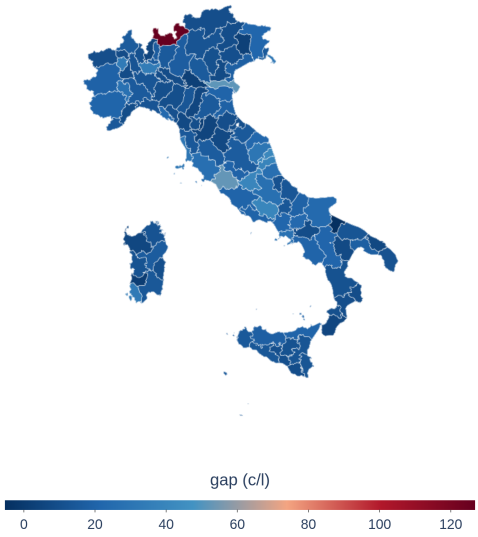


Figure 19: Non-adopters' mean gap to the provincial daily mean (c/l), pre-cap. Red = large gap.

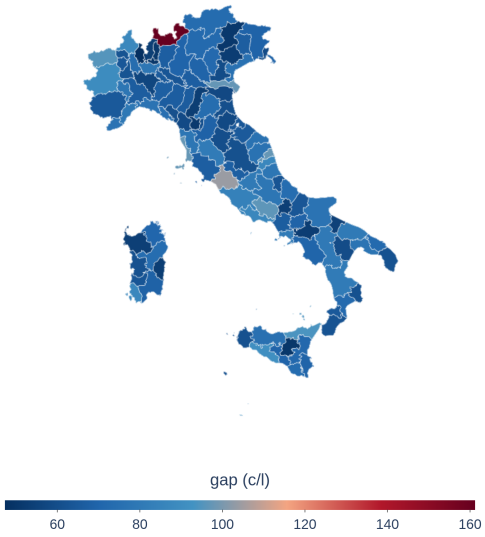


Figure 20: Non-adopters' mean gap to the provincial daily mean (c/l), post-cap (28 September – 11 October). Red = large gap.